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RedScrap: A Python Tool for Netnographic Data Collection from Reddit

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Software

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Summary

Netnography is a qualitative research method used to study online communities and digital interactions. Reddit, with its large user base and topic-specific subreddits, offers a rich source of publicly available content for exploring community discourse, opinions, and behaviors. However, collecting and organizing Reddit data for research purposes can be technically challenging, particularly for researchers unfamiliar with programming or API usage.

RedScrap is a lightweight and open-source Python tool designed to streamline Reddit data collection for netnographic and qualitative research. Built using the PRAW (Python Reddit API Wrapper) library, RedScrap allows users to extract structured data from Reddit threads or entire subreddits, including comments and metadata. The tool supports filtering by keyword and date range, enabling targeted data extraction across a wide range of research topics.

RedScrap provides both a command-line interface (CLI) and a user-friendly graphical interface (GUI), making it accessible to researchers with different levels of technical proficiency. The tool outputs data in standard formats (CSV and JSON), making it suitable for integration into qualitative coding workflows, such as thematic analysis, content analysis, and triangulation with other data sources.

Version **1.1.4** introduces two notable enhancements: an **Early Report** function, which generates a quick table of matching threads for preliminary scoping, and a **Scrape All Threads & Save** feature for one-click bulk extraction. These additions further streamline large-scale qualitative data collection while maintaining reproducibility.

RedScrap is designed for researchers in fields such as communication, sociology, digital ethnography, and public policy who seek efficient access to Reddit data. By lowering technical barriers and promoting reproducible practices, RedScrap contributes to the growing ecosystem of open-source tools that support ethical and scalable online research.

Statement of Need

Researchers in social sciences, digital ethnography, and communication studies increasingly rely on Reddit as a source of user-generated data to understand public opinion, discourse, and community interactions. Despite its value, Reddit data can be difficult to access in a structured form due to the technical complexity of APIs and the lack of user-friendly tools designed specifically for qualitative research workflows.

While existing Reddit scraping tools often prioritize large-scale quantitative data or are built for developers, there is a gap in tools that cater to qualitative researchers who need targeted, flexible, and ethical ways to collect Reddit content. RedScrap addresses this gap by offering an accessible, lightweight Python tool that enables researchers to extract Reddit comments and metadata with minimal setup.

RedScrap supports both code-based and GUI-based workflows, making it suitable for researchers with or without programming experience. It outputs data in standard formats (CSV and JSON), which are easily imported into qualitative analysis software for coding and interpretation.

By lowering the entry barrier to Reddit data collection and aligning with academic research practices, RedScrap meets the growing need for specialized tools that support qualitative and netnographic research in digital environments.

Installation

To install RedScrap and its dependencies, clone the repository and install the required Python packages using pip:

```
git clone https://github.com/yourusername/RedScrap.git
cd RedScrap
pip install -r requirements.txt
```

RedScrap requires Python 3.7 or later. It uses the PRAW library for accessing Reddit's API, along with standard packages for GUI and data export. Make sure to have a valid Reddit API client ID and secret, which can be obtained by registering an application at https://www.reddit.com/prefs/apps.

Usage

RedScrap can be used in two modes: via a Python script or through a simple graphical user interface (GUI). It allows users to extract Reddit comments and metadata for qualitative research by targeting specific posts, subreddits, keywords, and date ranges.

Python Script / CLI Usage

Here is a basic example using RedScrap in a Python script:

```
from redscrap import RedScrap

scraper = RedScrap()
scraper.scrape_post("https://www.reddit.com/r/malaysia/comments/xxxxx/sample_post/")
scraper.export_to_csv("output.csv")
```

You may also extract data from a subreddit by keyword:

```
scraper.scrape_subreddit("malaysia", keyword="legalization", limit=100)
```

Before using the script, ensure that your Reddit API credentials are configured (via `.env` or direct variable assignment inside the code, depending on your implementation).

GUI Usage

RedScrap includes a graphical interface that allows users to collect Reddit data without writing any code. To launch the GUI:

```
python gui.py
```

In the GUI: - Enter a subreddit name (e.g., `malaysia`) - Specify up to three comma-separated keywords (optional) - Define a date range (optional) - Click **Search Threads** to view matching threads - **Early Report**: Generate a CSV/PDF summary of all matching threads (titles, URLs, dates, upvotes, comment counts) for quick scoping - **Scrape Selected Thread & Save**: Export comments and metadata from a specific thread - **Scrape All Threads & Save**: Bulk export all comments and metadata from every matching thread - Output files are saved in CSV or JSON format, ready for analysis in software such as Excel, Atlas.ti, NVivo, or Python tools.

RedScrap GUI

Figure 1. RedScrap v1.1.4 graphical interface, showing the new "Early Report" and "Scrape All Threads & Save" buttons and the citation footer.

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Citation

If you use RedScrap in your research, please cite it as:

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References

Chereches, M., Finta, H., Prisada, R., & Rusu, A. (2024). Pharmacists' professional satisfaction and challenges: A netnographic analysis of Reddit and Facebook discussions. *Pharmacy*, 12, 1–33. https://doi.org/10.3390/pharmacy12050155

Chi, Y., & Chen, H. Y. (2023). Investigating substance use via Reddit: Systematic scoping review. *Journal of Medical Internet Research*, 25, e48905. https://doi.org/10.2196/48905

Jeacle, I. (2020). Navigating netnography: A guide for the accounting researcher. *Financial Accountability & Management*. https://doi.org/10.1111/faam.12237

Kozinets, R. V., & Gretzel, U. (2024). Netnography evolved: New contexts, scope, procedures and sensibilities. *Annals of Tourism Research*, 104, 103693. https://doi.org/10.1016/j.annals.2023.103693

Rocha-Silva, T., Nogueira, C., & Rodrigues, L. (2024). Passive data collection on Reddit: A practical approach. *Research Ethics*, 20(3), 453–470. https://doi.org/10.1177/17470161231210542

Strand, M. (2022). Attitudes towards disordered eating in the rock climbing community: A digital ethnography. *Journal of Eating Disorders*, 10(1), 96. https://doi.org/10.1186/s40337-022-00619-5

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